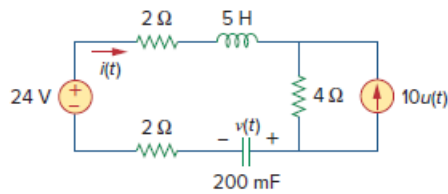


Problem

Obtain $v(t)$ and $i(t)$ for $t > 0$ in the circuit in Fig. 16.54.

Figure 16.54:



Step-by-step solution

Step 1 of 7

Refer to figure 16.54 in the textbook.

Consider that the capacitor acts like an open circuit and the inductor behaves like a short circuit for $t < 0$.

Draw the circuit for $t < 0$ as shown in figure 1.

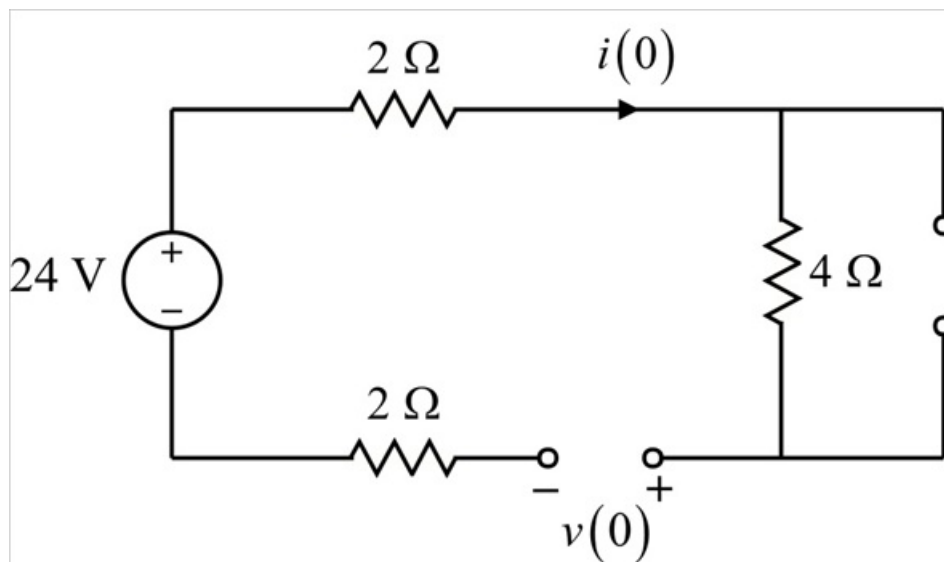


Figure 1

Step 2 of 7

Consider the circuit shown in figure 1.

Note that the capacitor doesn't allow sudden changes in voltage across it and the inductor doesn't allow sudden changes in the current passing through it.

Therefore,

$$i(0) = 0 \text{ A} \quad v(0) = 24 \text{ V}$$

Step 3 of 7

Redraw the circuit in s-domain with the initial values as shown in figure 2.

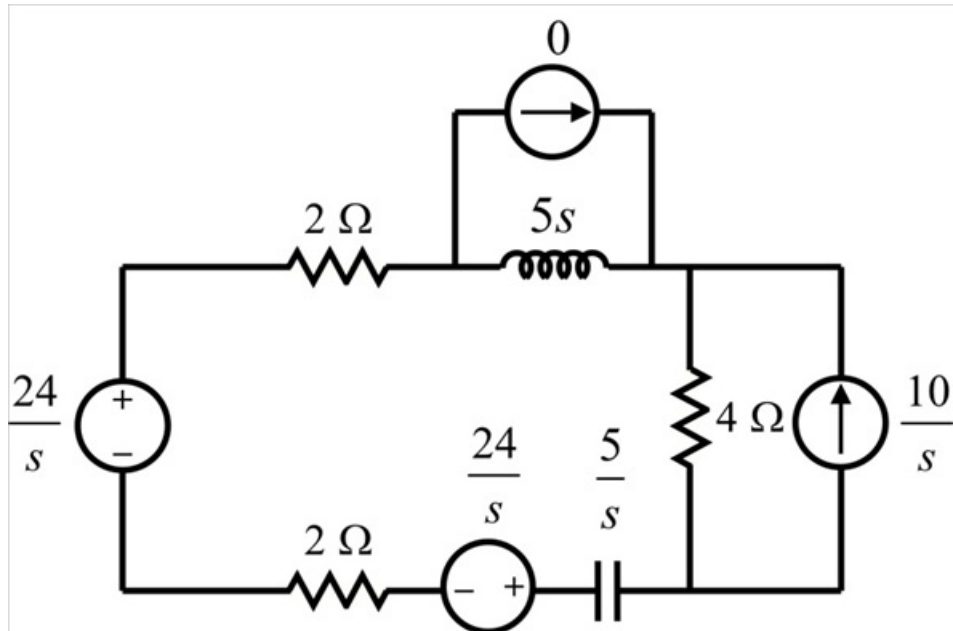


Figure 2

Step 4 of 7

Apply the source transformation technique to the combinations of impedances and current sources of the circuit shown in figure 2.

Redraw the circuit as shown in figure 3.

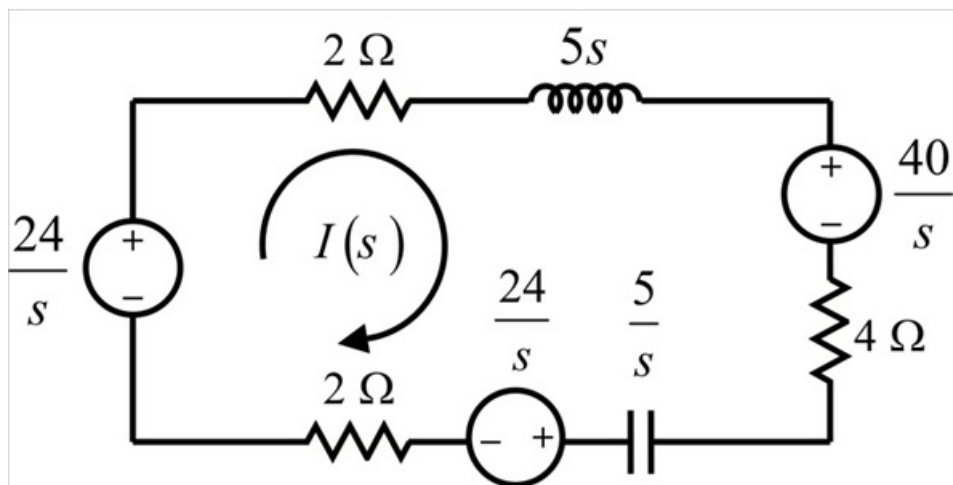


Figure 3

Step 5 of 7

Apply the Kirchhoff's voltage law to the circuit shown in figure 3.

$$-\frac{24}{s} + 2I(s) + 5sI(s) + \frac{40}{s} + 4I(s) + \frac{5}{s}I(s) + 2I(s) + \frac{24}{s} = 0$$

$$8I(s) + 5sI(s) + \frac{5}{s}I(s) + \frac{40}{s} = 0$$

$$I(s) \left(8 + 5s + \frac{5}{s} \right) = -\frac{40}{s}$$

$$I(s) \left(\frac{5s^2 + 8s + 5}{s} \right) = \frac{-40}{s}$$

$$I(s) = \frac{-40}{5s^2 + 8s + 5}$$

$$= \frac{-8}{s^2 + \frac{8}{5}s + 1}$$

$$= \frac{-8}{s^2 + 2\left(\frac{4}{5}\right)s + \left(\frac{4}{5}\right)^2 + \left[1 - \left(\frac{4}{5}\right)^2\right]}$$

$$= \frac{-8}{\left(s + \frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2}$$

Step 6 of 7

Therefore, the current flowing through the inductor is,

$$I(s) = \frac{-8}{\left(s + \frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2}$$

Apply the inverse Laplace transform.

$$\begin{aligned} i(t) &= -8e^{-\frac{4}{5}t} \sin\left(\frac{3}{5}t\right) u(t) \\ &= -8e^{-0.8t} \cos(0.6t + 90^\circ) u(t) \end{aligned}$$

Therefore, the current flowing through the inductor is $\boxed{-8e^{-0.8t} \cos(0.6t + 90^\circ) u(t) \text{ A}}.$

Step 7 of 7

The voltage across the capacitor in s-domain is,

$$\begin{aligned}
 V(s) &= \frac{24}{s} + \frac{5}{s} I(s) \\
 &= \frac{24}{s} + \frac{5}{s} I(s) \\
 &= \frac{24}{s} - \frac{5}{s} \left[\frac{40}{5s^2 + 8s + 5} \right] \\
 &= \frac{24}{s} - \frac{40}{s} + \frac{40 \left(s + \frac{8}{5} \right)}{s^2 + \frac{8}{5}s + 1} \\
 &= -\frac{16}{s} + \frac{40 \left(s + \frac{4}{5} + \frac{4}{5} \right)}{\left(s + \frac{4}{5} \right)^2 + \left(\frac{3}{5} \right)^2} \\
 &= -\frac{16}{s} + \frac{40 \left(s + \frac{4}{5} \right)}{\left(s + \frac{4}{5} \right)^2 + \left(\frac{3}{5} \right)^2} + \frac{(53.33) \left(\frac{3}{5} \right)}{\left(s + \frac{4}{5} \right)^2 + \left(\frac{3}{5} \right)^2}
 \end{aligned}$$

Apply the inverse Laplace transform.

$$\begin{aligned}
 v(t) &= -16u(t) + 40e^{-\frac{4}{5}t} \cos\left(\frac{3}{5}t\right)u(t) + 53.33e^{-\frac{4}{5}t} \sin\left(\frac{3}{5}t\right)u(t) \\
 &= -16u(t) + \frac{40}{0.6}e^{-\frac{4}{5}t} \cos\left(\frac{3}{5}t\right) \cos(53.13^\circ)u(t) + \frac{53.33}{0.8}e^{-\frac{4}{5}t} \sin\left(\frac{3}{5}t\right) \sin(53.13^\circ)u(t) \\
 &= -16 + 66.67e^{-\frac{4}{5}t} \left[\cos\left(\frac{3}{5}t\right) \cos(53.13^\circ) + \sin\left(\frac{3}{5}t\right) \sin(53.13^\circ) \right] u(t) \\
 &= \left[-16 + 66.67e^{-0.8t} \cos(0.6t - 53.13^\circ) \right] u(t)
 \end{aligned}$$

Therefore, the voltage across the capacitor for $t > 0$ is

$$\boxed{\left[-16 + 66.67e^{-0.8t} \cos(0.6t - 53.13^\circ) \right] u(t)}.$$